STAT 559 – Note 01
Measure Spaces and Extension: Foundations

Edited transcription of handwritten lecture notes

These notes transcribe and edit the indicated handwritten PDF. Source-page markers preserve traceability. Yellow boxes retain original TODOs; red boxes explain mathematical corrections; blue boxes add explanations or proof details. The reference is *stats310notes.pdf* (provided in `notes/`); references below use its printed section numbers.

Source page(s): 1–3

1 Why measure theory?

The central question is how to define length on $\mathbb{R}$, area on $\mathbb{R}^2$, and volume on $\mathbb{R}^3$, especially for complicated sets. The notes connect Borel's measurable spaces, set theory, and probability. Let Ω be a set with elements $\omega \in \Omega$. A *sigma-algebra* (or sigma-field) $\mathcal{F} \subseteq 2^\Omega$ satisfies

$$\varnothing \in \mathcal{F}, \tag{1}$$

$$A \in \mathcal{F} \Longrightarrow A^c = \Omega \setminus A \in \mathcal{F}, \tag{2}$$

$$A_1, A_2, \ldots \in \mathcal{F} \Longrightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}. \tag{3}$$

The pair $(\Omega, \mathcal{F})$ is a *measurable space*. A *measure* on it is a function $\mu : \mathcal{F} \to [0, \infty]$ such that $\mu(\varnothing) = 0$ and, for pairwise disjoint measurable sets,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n). \tag{4}$$

The triple $(\Omega, \mathcal{F}, \mu)$ is a *measure space*.

Original TODO: Why restrict the sets that can be measured?

The handwritten illustration mentions the Banach–Tarski ("sun–pea") paradox: finitely many pieces of a three-dimensional ball can be rearranged by rigid motions into two balls of the original size. The pieces cannot all be measurable for the usual volume. This motivates choosing a suitable sigma-algebra rather than requiring every subset to have geometric volume. The paradox is motivational here, not proved in the supplied reference.

Correction: Degenerate measures and geometric volume

The source suggests allowing only zero or infinity as an alternative. The rule $\mu(\varnothing) = 0$ and $\mu(A) = \infty$ for every nonempty A is indeed a measure, but it does not extend ordinary length or volume. A useful geometric measure must retain the prescribed values on intervals or boxes.

Source page(s): 4–6

2 Generated sigma-algebras and topology

A sigma-algebra contains $\Omega = \varnothing^c$ and is closed under countable intersections, by De Morgan's law. The intersection of any nonempty family of sigma-algebras on the same Ω is again a sigma-algebra. For any class $\mathcal{A} \subseteq 2^\Omega$, the *sigma-algebra generated by* $\mathcal{A}$ is

$$\sigma(\mathcal{A}) = \bigcap \{\mathcal{G} : \mathcal{G} \text{ is a sigma-algebra on } \Omega,\ \mathcal{A} \subseteq \mathcal{G}\}. \tag{5}$$

The family being intersected is nonempty because it contains 2^Ω. The intersection is a sigma-algebra, contains $\mathcal{A}$, and is contained in every sigma-algebra containing $\mathcal{A}$. This proves existence and minimality. A *topology* $\tau \subseteq 2^\Omega$ contains $\varnothing, \Omega$ and is closed under arbitrary unions and finite intersections. Its members are open sets. The Borel sigma-algebra is $\mathcal{B}(\Omega) = \sigma(\tau)$.

Correction: Collections versus elements

The source calls a topology a subset of Ω. It is a collection of subsets, hence a subset of 2^Ω. Likewise, the family of sigma-algebras in the preceding proof is an indexed family; it need not be countably enumerable.

Source page(s): 7–8

On $\mathbb{R}$ with its usual topology, $\mathcal{B}(\mathbb{R})$ is generated by all open subsets. Measurable sets in a probability space are called *events*. A probability space is a measure space with $\mu(\Omega) = 1$.

3 Basic properties of measures

If $A \subseteq B$ are measurable, disjoint additivity gives

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A). \tag{6}$$

For arbitrary measurable A_n, the *union bound* is

$$\mu\left(\bigcup_{n\geq 1} A_n\right) \leq \sum_{n\geq 1} \mu(A_n). \tag{7}$$

For example, disjointify the union using $D_1 = A_1$ and $D_n = A_n \setminus \bigcup_{j<n} A_j$; then apply additivity and $D_n \subseteq A_n$. If $A_n \uparrow A$, then

$$\mu(A) = \lim_{n\to\infty} \mu(A_n). \tag{8}$$

Indeed, apply countable additivity to $A_1, A_2 \setminus A_1, \ldots$. If $A_n \downarrow A$ and $\mu(A_1) < \infty$, then

$$\mu(A) = \lim_{n\to\infty} \mu(A_n). \tag{9}$$

Apply continuity from below to $A_1 \setminus A_n$ and subtract from $\mu(A_1)$.

Original TODO: Why is finiteness needed for continuity from above?

The source assumes the stronger condition $\mu(\Omega) < \infty$. It suffices that one set of the decreasing sequence has finite measure. Without this, take Lebesgue measure and $A_n = [n, \infty)$: every $\mu(A_n) = \infty$, while $\bigcap_n A_n = \varnothing$ has measure zero. See the reference, Section 1.2, continuity of measures.

Source page(s): 9–10

4 Algebras and approximation

An *algebra* $\mathcal{A}$ contains $\varnothing$, is closed under complements, and is closed under finite unions.

Correction: Finite union in the definition

The source writes an infinite upper limit next to the words "finite union." The intended condition is $\bigcup_{j=1}^{n} A_j \in \mathcal{A}$ for every finite n.

Approximation theorem. Suppose $\mu(\Omega) < \infty$ and $\mathcal{F} = \sigma(\mathcal{A})$, where $\mathcal{A}$ is an algebra. For every $A \in \mathcal{F}$ and $\varepsilon > 0$ there is $B \in \mathcal{A}$ with

$$\mu(A \triangle B) < \varepsilon, \qquad A \triangle B = (A \setminus B) \cup (B \setminus A). \tag{10}$$

The notes state this for a probability space. Define $\mathcal{G}$ as the measurable sets with this approximation property. Then $\mathcal{A} \subseteq \mathcal{G} \subseteq \mathcal{F}$. Complements preserve symmetric differences, so $\mathcal{G}$ is closed under complements. For $A_n \in \mathcal{G}$, put $A = \bigcup_n A_n$ and $C_N = \bigcup_{n\le N} A_n$. By continuity and finiteness, choose N with $\mu(A \setminus C_N) < \varepsilon/2$. Choose $B_n \in \mathcal{A}$ for $n \le N$ with $\mu(A_n \triangle B_n) < \varepsilon/(2N)$. Then $B = \bigcup_{n\le N} B_n \in \mathcal{A}$ and

$$\mu(A \triangle B) \le \mu(A \setminus C_N) + \sum_{n=1}^{N} \mu(A_n \triangle B_n) < \varepsilon. \tag{11}$$

Thus $\mathcal{G}$ is a sigma-algebra containing $\mathcal{A}$, so $\mathcal{G} = \mathcal{F}$.

Supplement: Completing the proof sketch

The handwritten proof gives the sigma-algebra strategy but leaves its closure under countable unions to be checked. The finite truncation argument above supplies that step. Compare the reference Theorem 1.2.6.

Source page(s): 11–12

5 Constructing length and extending a measure

The desired interval values are $\mu((a,b]) = b - a$ for real $a \le b$. Begin with simple intervals, prescribe their lengths, and extend to their generated sigma-algebra.

Correction: The starting class and uniqueness hypotheses

The source alternates between open and half-open intervals, and calls the initial triple a measure space before a sigma-algebra has been constructed. A consistent construction starts with half-open intervals (and the empty set), extends length to their finite disjoint unions, and obtains a premeasure on that algebra. Infinite endpoints need appropriate ray conventions. Carathéodory extension then gives a measure on the generated Borel sigma-algebra. Uniqueness requires an appropriate sigma-finiteness assumption; it is not automatic for an arbitrary initial set function. Its completion is Lebesgue measure on the Lebesgue sigma-algebra. See reference Sections 1.5–1.7.

Source page(s): 13–14

6 The pi–lambda theorem and uniqueness

A *pi-system* $\mathcal{P}$ is a class closed under finite nonempty intersections. A *lambda-system* $\mathcal{L}$ contains Ω, is closed under complements, and is closed under countable disjoint unions. A lambda-system that is also a pi-system is a sigma-algebra: intersections and complements give finite unions and differences; disjointification then gives countable unions. **Dynkin's pi–lambda theorem.** If $\mathcal{P}$ is a pi-system and $\mathcal{L}$ is a lambda-system with $\mathcal{P} \subseteq \mathcal{L}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$. **Uniqueness theorem.** Let μ_1, μ_2 be measures on $\sigma(\mathcal{P})$ agreeing on a pi-system $\mathcal{P}$. Suppose there are $E_n \in \mathcal{P}$ with $E_n \uparrow \Omega$ and $\mu_1(E_n) = \mu_2(E_n) < \infty$. Then $\mu_1 = \mu_2$ on $\sigma(\mathcal{P})$.

Source page(s): 15–16

Proof. Fix $E \in \mathcal{P}$ of finite common measure, and set

$$\mathcal{L}_E = \{B \in \sigma(\mathcal{P}) : \mu_1(E \cap B) = \mu_2(E \cap B)\}. \tag{12}$$

It contains Ω. If $B \in \mathcal{L}_E$, finite subtraction gives

$$\mu_1(E \cap B^c) = \mu_1(E) - \mu_1(E \cap B) \tag{13}$$

$$= \mu_2(E) - \mu_2(E \cap B) = \mu_2(E \cap B^c). \tag{14}$$

For disjoint $B_n \in \mathcal{L}_E$,

$$\mu_1\left(E \cap \bigcup_n B_n\right) = \sum_n \mu_1(E \cap B_n) \tag{15}$$

$$= \sum_n \mu_2(E \cap B_n) = \mu_2\left(E \cap \bigcup_n B_n\right). \tag{16}$$

Thus $\mathcal{L}_E$ is a lambda-system. Since $E \cap B \in \mathcal{P}$ for $B \in \mathcal{P}$, it contains $\mathcal{P}$. Dynkin's theorem gives $\mathcal{L}_E = \sigma(\mathcal{P})$. Therefore, for all $B \in \sigma(\mathcal{P})$, continuity from below yields

$$\mu_1(B) = \lim_n \mu_1(E_n \cap B) = \lim_n \mu_2(E_n \cap B) = \mu_2(B). \tag{17}$$

Source page(s): 17

7 Carathéodory's roadmap

First extend a premeasure from an algebra to an outer measure on all subsets. Next restrict that outer measure to its Carathéodory-measurable sets, which form a sigma-algebra. Finally restrict to the sigma-algebra generated by the original algebra:

$$(\mathcal{A}, \mu_0) \longrightarrow (2^\Omega, \mu^*) \longrightarrow (\mathcal{M}_{\mu^*}, \mu^*|_{\mathcal{M}_{\mu^*}}) \longrightarrow (\sigma(\mathcal{A}), \mu^*|_{\sigma(\mathcal{A})}). \tag{18}$$

The first object in the middle carries an outer measure, not in general a measure. The next note supplies the two Carathéodory theorems indicated in the source diagram.